Sharp denominator corrections for Brown–Zudilin cellular linear forms in $\zeta(5)$: measurements, a conjecture, and a lattice mechanism

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July 2026

Abstract

Brown and Zudilin [arXiv:2210.03391] constructed effective rational approximations $I'(\mathbf{a}\cdot n) = Q_n\zeta(5) - P_n$ to $\zeta(5)$ from cellular integrals on $\overline{\mathcal{M}}_{0,8}$, together with experimental integrality statements for the coefficients. By recovering the exact rational coefficients $P_n, \hat{P}_n$ at more than 200 parameter points via certified integer-relation computations, we show that the published integrality claims are false as stated—their own printed values violate them—and that the correct statements carry a correction factor of exactly $12 = 2^2 \cdot 3$ (respectively 2 for the $\zeta(3)$-companion), which we conjecture to be sharp and verify in every measured cell. Unlike the familiar correction factors in the classical literature (Apéry, Rhin–Viola, Krattenthaler–Rivoal), which we check are uniformly *removable*, this correction is *attained.* We propose a mechanism: the coefficients are periods against a non-primitive integral Betti class produced by the construction's motivic elimination of $\zeta(2)$, so the correction is a lattice index bounded by the Bernoulli constant 24 arising from $\zeta(2) = -(2\pi i)^2/24$; an elementary elimination-cost lemma, a conditional theorem, and a heuristic localization in the totally symmetric case (the factor 12 appearing as a doubled sine-kernel Laurent coefficient $\frac{1}{6}$; all symmetric-case laws verified exactly for $n \leq 8$) support this. As by-products we determine the supremum of the Brown–Zudilin "worthiness" exponent over the full parameter cone, verify that the leading factor 2 in a denominator conjecture of Krattenthaler–Rivoal is unnecessary for $n \leq 5$, and document an orbit-coherent boundary-prime anomaly. In the totally symmetric case we prove one law unconditionally: reducing to Zudilin's 2002 sequences, the corrected integrality is equivalent to the central-binomial divisibility $\binom{2n}{n} \mid 6A_n$, and we prove it for *all* n at every prime $p \in (n, 2n]$—the primes invisible to every d_n-denominator estimate—by a new Frobenius-certificate argument: the single polynomial $(k^p - k)^2$ realizes, over $\mathbb{F}_p$, the $n+1$ simultaneous local jet conditions that admit no low-degree solution over $\mathbb{Q}$, and a residue-sum argument on $\mathbb{P}^1$ does the rest. The primes $p \leq n$ of that divisibility remain open. All measured denominator slack is bounded; no measured direction offers exponential arithmetic gain.

1 Introduction

For the linear forms of Apéry-type used in irrationality proofs, the arithmetic of the rational coefficients is traditionally expressed by *inclusions*: statements of the shape $D_n \cdot (\text{coefficient}) \in$

*Draft of 16 July 2026. Computations, conjectures and proofs in this paper were produced in a human–AI collaboration; the verification code and complete datasets are available in the accompanying repository. Correspondence: [address].

$\mathbb{Z}$ for explicit denominators D_n built from $d_n = \mathrm{lcm}(1,\ldots,n)$. These inclusions are proved (or conjectured) from term-by-term analysis, and the question of their *sharpness*—whether the true denominators actually attain D_n—is rarely addressed. Proven *upper* bounds in the cellular program do exist (most recently Eyre–Muller–Nieto for the Catalan constant [15], whose Thm. 8.3.1 clears denominators by an explicit factor, with no motivic interpretation claimed); what has, to our knowledge, never been done is the systematic *measurement* of true denominators — sharp, per-prime, and interpreted as lattice data.

This paper does so for the family of cellular integrals $I(\mathbf{a})$ on $\overline{\mathcal{M}}_{0,8}$ introduced by Brown and Zudilin [1], which produce simultaneous rational approximations $I' = Q\zeta(5) - P$, $I'' = Q\zeta(3) - \hat{P}$ by a motivic elimination of the parasitic $\zeta(2)$. Our findings:

1. The integrality statements printed in [1] (their experimental inclusions for I', and the totally symmetric display $d_n^5 P_n \in \mathbb{Z}$) are **false as stated**, already against the paper's own printed values (§3).

2. The corrected statements carry a factor of exactly 12 (resp. 2), and this factor is **sharp**: it is attained in 46 of 164 systematically sampled cells at the prime 2 and 30 cells at the prime 3, and never exceeded in any of our more than 200 certified cells (§4). One isolated, orbit-coherent, non-persistent exception at a boundary prime $p = 7$ is documented in §5.

3. By contrast, the analogous correction factors appearing in the classical literature are uniformly *removable*: we verify $d_n^3 a_n \in \mathbb{Z}$ for Apéry's $\zeta(3)$ forms for $n \leq 50$ (sharpening the ubiquitous "$2d_n^3$"), and we verify that the leading factor 2 retained even in the *conjectured* sharp denominator law of Krattenthaler–Rivoal [5, eq. (17.1)] for Zudilin's $\zeta(5),\ldots,\zeta(11)$ forms is unnecessary for $n \leq 5$ (§8)—apparently the first direct computation of those coefficients.

4. We propose a mechanism (§6): the elimination of $\zeta(2)$ forces the linear form to be the period of a *non-primitive* integral Betti class, and the identity $\zeta(2) = -(2\pi i)^2/24$ prices any integral elimination at a lattice index dividing 24. An elementary lemma makes this precise; the measured constants 12 and 2 then *measure* 2-adic indices of the integral Betti lattice of the underlying rank-3 mixed Tate motive. In the totally symmetric case the excess is localized (heuristically) in the doubled Laurent coefficient $\frac{1}{6}$ of the reciprocal-sine kernel; no $p \geq 5$ excess is observed through $n = 8$, though the support statement remains conjectural (§7).

5. All measured denominator slack (cells where the true denominator is *smaller* than predicted) is bounded along every measured direction, including growth series $n \leq 8$; consequently the construction's "worthiness" budget receives no hidden arithmetic subsidy. We also determine $\sup \gamma = 0.86597135\ldots$ over the full 7-dimensional projective parameter cone—attained exactly at the authors' published point, whose symmetric coordinates form an arithmetic progression (§9).

Throughout, "verified" means: exact integers Q_n from the closed binomial double sum of [1]; values of $I(\mathbf{a}\cdot n)$ computed to a certified precision (hundreds of digits) from the 1-dimensional integral of a product of two ${}_3F_2$'s given in [1], evaluated by an exact partial-fraction/polylogarithm method; and $P_n, \hat{P}_n$ recovered by PSLQ over the basis $\{1, \zeta(2)\}$ with residual certification, precision-stability checks, and exact agreement with all rational values printed in [1] (§2).

2 The family and the measurement pipeline

We follow the notation of [1]. For admissible integer parameters $\mathbf{a} = (a_1, \ldots, a_8)$ (the 17 convergence forms nonnegative; we require all 28 forms of the multiset $\mathbf{h}(\mathbf{a})$ nonnegative so that the full symmetry orbit is admissible), the cellular integral decomposes as

$$I(\mathbf{a}) \;=\; 2Q\,(\zeta(5) + 2\zeta(3)\zeta(2)) \;-\; 4\hat{P}\,\zeta(2) \;-\; 2P, \qquad Q \in \mathbb{Z},\; P, \hat{P} \in \mathbb{Q},$$

and the object of interest is $I' = Q\zeta(5) - P$. The group $\mathfrak{G} \cong \Sigma_7$ acts on normalized integrals through the symmetric parameters $s_0; s_1, \ldots, s_7$, permuting the 28-element multiset $\mathbf{h}$, with $I(\mathbf{a})/\prod_{i\in\mathcal{F}} h_i!$ invariant for the 17-element index set $\mathcal{F}$ of [1].

Exact Q. $Q(\mathbf{a}\cdot n)$ is computed exactly from the double binomial sum [1, eq. (sumQ)].

Certified I. $J(\mathbf{p};\mathbf{q}) = I(\mathbf{a})$ reduces [1] to a 1-dimensional integral of a product of two ${}_3F_2$'s. For integer parameters each ${}_3F_2$ coefficient is a proper rational function of the summation index; its exact partial fractions over $\mathbb{Q}$ convert every evaluation into finitely many polylogarithms $\mathrm{Li}_m(z)$, $z \le 0$, with rational weights (analytic continuation built in). Working precision is raised until two runs agree; tanh–sinh quadrature errors are certified against the target precision. On validation, this evaluator agrees with the direct (slow) hypergeometric path to all compared digits (≥ 40) and is $\sim 57\times$ faster on production cells.

Recovery. With ζ-values to matching precision, $x = (2Q\zeta(5) + 4Q\zeta(3)\zeta(2) - I)/2 = P + 2\hat{P}\zeta(2)$, and PSLQ on $(x, 1, \zeta(2))$ returns $P, \hat{P}$ as exact rationals. Certification: the relation residual must vanish to half the working precision; the recovery must be stable under a +80-digit rerun; and the pipeline must reproduce the exact printed values of [1] in the totally symmetric case,

$$Q_1 = 21, \quad P_1 = \tfrac{87}{4}, \quad \hat{P}_1 = \tfrac{101}{4}, \qquad Q_2 = 2989, \quad P_2 = \tfrac{1190161}{384}, \quad \hat{P}_2 = \tfrac{344923}{96},$$

which it does. An incidental end-to-end validation appears in §5: two independent recoveries at distinct parameter points agree with the exact factorial-ratio identity $I'(g\mathbf{a}) = I'(\mathbf{a}) \cdot \prod h_i(\mathbf{a})!/\prod h_i(g\mathbf{a})!$ through a discrepancy of 7^2.

3 The published inclusions are false as stated

Brown–Zudilin state (experimentally) the inclusion $d_{m_1 n} \cdots d_{m_5 n}\, I'(\mathbf{a}\cdot n) \in \mathbb{Z}\zeta(5) + \mathbb{Z}$, where $m_1 \ge \cdots \ge m_5$ are the five largest entries of $\mathbf{h}(\mathbf{a})$, together with a sharpening by the factor $\Phi_n = \prod_{p > \sqrt{m_1 n}} p^{\nu_p}$, and, in the totally symmetric case, the display $Q_n,\ d_n^2 d_{2n}\hat{P}_n,\ d_n^5 P_n \in \mathbb{Z}$.

Already at $n = 2$ in the totally symmetric case, with the paper's own printed values: $d_2^5 P_2 = 2^5 \cdot \frac{1190161}{384} = \frac{1190161}{12} \notin \mathbb{Z}$ (off by $2^2 \cdot 3$), and $d_2^2 d_4 \hat{P}_2 = 48 \cdot \frac{344923}{96} = \frac{344923}{2} \notin \mathbb{Z}$ (off by 2). The numerators are coprime to 6, so the failures are exact. Our recovered values agree with the printed ones, independently. Analogous failures occur at 3 of 19 small interior points in our first systematic sample, always 2-adically. The corrected statements are the subject of the next section.

4 The sharp correction: data and conjecture

Extend Brown–Zudilin's sharpening to every prime (legitimate, since the factorial-ratio identity above is exact p-adically):

$$D_\nu(\mathbf{a}, n) = \frac{\prod_{i=1}^5 d_{m_i n}}{\Phi_n^*}, \qquad \Phi_n^* = \prod_{\text{all } p} p^{\nu_p}, \qquad \nu_p = \max_{g \in \mathfrak{G}} \operatorname{ord}_p \frac{\prod_{i\in\mathcal{F}}(h_i(\mathbf{a})n)!}{\prod_{i\in\mathcal{F}}(h_i(g\mathbf{a})n)!}.$$

Conjecture 4.1 (robust form)**.** *For all admissible* $\mathbf{a}$ *and* $n \geq 1$*, and every prime* p*,*

$$\operatorname{ord}_p \operatorname{den}(P_n) \;\leq\; \operatorname{ord}_p D_\nu(\mathbf{a}, n) \;+\; \begin{cases} 2, & p = 2, \\ 1, & p = 3, \\ 1, & p \geq 5, \end{cases}$$

with the ceilings at 2 *and* 3 *attained, the* $p \geq 5$ *term nonzero only at boundary primes* $p = m_1$*, and, in the* ***strong form****, vanishing for* $n \geq 2$*.*

We emphasize the epistemic status: both forms are falsifiable by a single future cell, and the boundedness of the $p \geq 5$ term—like every sharpness statement here—rests on finite evidence only. We know of no structural argument excluding infinitely many boundary-prime fluctuations; §5 documents the one observed orbit and the failed attempt to make it recur.

Evidence. Over 200 certified cells (systematic random interior points with entries up to 7, growth series at selected points with $n \leq 8$, and the authors' record point): the 2-adic excess distribution over the first 164 cells is $\{-6{:}2,\ -5{:}1,\ -4{:}4,\ -3{:}9,\ -2{:}9,\ -1{:}20,\ 0{:}33,\ +1{:}40,\ +2{:}46\}$ — ceiling $+2$, attained 46 times, never exceeded; the 3-adic distribution is $\{-4{:}1,\ -3{:}3,\ -2{:}8,\ -1{:}43,\ 0{:}133,\ +1{:}30\}$ — ceiling $+1$, attained, never exceeded. In the totally symmetric case the 2-adic ceiling is attained at every computed n ($n = 1, \ldots, 5$ and 8): $\operatorname{den}(P_n) = 4 \cdot 2^{\operatorname{ord}_2 d_n^5} \cdot$ (odd part), and $\operatorname{den}(\hat{P}_n) = 2 \cdot 2^{\operatorname{ord}_2 d_n^2 d_{2n}} \cdot$ (odd part) for $n \leq 5$. The odd parts show only slack (the 3-adic prediction over-pays at most points, which is why the 3-part of the correction is visible only where the 3-adic prediction is tight).

Growth. Along every measured series the excess/slack trajectory is bounded: e.g. the largest observed slack point $\mathbf{a} = (4, 3, 4, 4, 3, 2, 4, 2)$ has 2-adic slack $6, 5, 4, 5, 3$ bits at $n = 1, \ldots, 5$; points with attained ceiling at $n = 1, 2$ can relax into slack at $n = 3$. The conjectured bound—not any pointwise formula—is the stable object.

5 A boundary-prime anomaly, and orbit coherence

Exactly one sampled orbit violates the $p \geq 5$ clause: at $\mathbf{a} = (6, 2, 6, 1, 6, 4, 5, 5)$, $n = 1$ (where $m_1 = 7$ is prime and $\nu_7 = 1$), $\operatorname{den}(P_1) = 2^7 \cdot 3^4 \cdot 5^2 \cdot 7$ exceeds the ν-sharpened prediction by one factor of 7. Since the factorial-ratio identity is exact, this *predicts* that the maximizing orbit partner $\mathbf{a}' = (6, 2, 5, 2, 4, 6, 6, 5)$ must violate even the raw (ν-free) law; direct audit confirms $\operatorname{den}(P_1(\mathbf{a}')) = 2^6 \cdot 3^4 \cdot 5^2 \cdot 7^2$ against a d-product paying a single 7. Excesses are thus *orbit-coherent*: the correct invariant is an orbit-level anomaly divisor.

In our samples the effect is rare and non-persistent: a targeted scan of 40 fresh points satisfying the same trigger condition ($m_1 \in \{5, 7, 11, 13\}$ prime, $\nu_{m_1} \geq 1$) produced *no* further instance, and the anomalous orbit itself relaxes to slack (-2 at $p = 7$) at $n = 2$, at both partners coherently. We stress that this falsifies the naive "$\{2, 3\}$ only" clause outright, and that finite sampling cannot exclude infinitely many such violations across the cone or along n; the robust form of Conjecture 4.1 prices them at one unit at the boundary prime, and even that bound is an empirical reading, not a theorem. Characterizing the boundary-prime effect exactly is open, and we regard it as the sharpest known probe of what the $\mathfrak{S}$-symmetrized inclusion machinery misses.

6 The mechanism

Lemma 6.1 (elimination cost)**.** *Let e_0, e_2 span a 2-dimensional $\mathbb{Q}$-vector space and let L be a lattice in the dual with basis γ_1, γ_2 pairing as $\langle \gamma_1, (e_0, e_2)\rangle = (1, \zeta(2))$, $\langle \gamma_2, (e_0, e_2)\rangle = (0, (2\pi i)^2)$. Every nonzero $\delta \in L$ with $\langle \delta, e_2\rangle = 0$ is an integer multiple of $\delta_0 = 24\gamma_1 + \gamma_2$; in particular $\langle \delta, e_0\rangle \in 24\mathbb{Z}$.*

Proof. $a\zeta(2) + b(2\pi i)^2 = 0$ and $\zeta(2)/(2\pi i)^2 = -\frac{1}{24}$ force $b = a/24$, so $24 \mid a$. □

If the true integral lattice is finer, with $\gamma_2/2^k$ integral, the cost drops to $24/2^k$. The linear form I' arises exactly by such an elimination: $\mathrm{Ext}^1_{MT(\mathbb{Z})_{\mathbb{Q}}}(\mathbb{Q}(0), \mathbb{Q}(2)) = 0$ [7] splits the weight-4 part rationally, and the identity $\zeta^{\mathfrak{m}}(2) = -(\mathbb{L}^{\mathfrak{m}})^2/24$ is available motivically [8, §2.2]. The systematic ($\mathbb{Q}$-linear) theory of such extension structures for zeta-value linear forms is due to Dupont [13, §1.2, Thms. 1.3–1.4], whose *odd zeta motive* realizes all the relevant extensions; we note that the Brown–Zudilin motive here, having the product period $\zeta(5) + 2\zeta(3)\zeta(2)$ in top weight, is provably *not* a subquotient of Dupont's motive, and the precise relationship is open (raised in [13]). Everything integral — lattices, denominators, indices — is outside the scope of [13], which is $\mathbb{Q}$-linear throughout. Under two explicitly stated hypotheses—(H1) the integrand's de Rham coefficients are D_ν-controlled (the integral form of Brown–Zudilin's own sketched Barnes analysis), and (H2) a normalization of the integral Betti lattice—one obtains $24\, D_\nu\, I' \in \mathbb{Z}\zeta(5) + \mathbb{Z}$, with the constant improving by the lattice index. The measured sharp constants then *measure* that index: $12 = 24/2$ for P (equivalently, cost 2 on the natural coefficient $2P$), and 2 for $\hat{P}$.

6.1 The lattice index at the symmetric point: locating the factor 2

At the totally symmetric point the ambiguity left open by the measurement — the constant $12 = 24/2$ fixes only the *product* of the Betti lattice index and the de Rham normalization of the integrand — can be resolved by explicit computation. Full details, scripts, and a convention-risk register appear in the companion notes (`H2_LATTICE.md`); we summarize the three inputs and the conclusion.

Setup. The singular divisor of the symmetric integrand is $A = \{\delta_{24}, \delta_{14}, \delta_{57}, \delta_{35}, \delta_{36}\}$ (matching the exponent subscripts of the family's defining integral), B is the associahedron boundary ($A \cap B = \emptyset$), and the rank-one piece $\mathrm{gr}^W_4 \cong \mathbb{Q}(-2)$ of the motive is carried by the seven transverse codimension-2 strata inside A. Ambient ranks were cross-checked against Keel's formulas ($b_2(\overline{\mathcal{M}}_{0,8}) = 99$; Betti string $1, 99, 626, 626, 99, 1$), and the involution $\mathfrak{i}_1$ acts as $+1$ on gr^W_4 (forced by functoriality and $\hat{P} \neq 0$), which already pins any lattice anomaly to the prime 2: no order-3 symmetry is available.

Input 1 (de Rham side, exact). The iterated residues of the symmetric log-form ω along all seven strata, computed exactly with tracked orientation signs, are all equal to ± 1. Hence ω restricts to a *primitive* integral cellular form on each stratum: the integral de Rham class in

gr_4^W carries no factor of 2 (nor of 3). This refutes the alternative reading in which the measured 2 is a de Rham normalization effect.

Input 2 (arrangement combinatorics, exact). The incidence chain complex of the polar arrangement (5 divisors, 7 transverse codimension-2 strata, 2 codimension-3 triples), assembled as integer boundary matrices, has Smith normal form with all elementary divisors 1 ($H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}$, $H_2 = 0$): the arrangement contributes no 2-torsion and no divisibility.

Input 3 (measurement). ord_2 of the clearing constant is 1 on the natural coefficient (§4).

Conclusion. Valuations add along the period pairing, so $v_2(\text{Betti index}) = v_2(\text{measured}) - v_2(\text{de Rham}) = 1 - 0 = 1$: **the factor 2 is a genuine refinement of the integral Betti lattice** ($\gamma_2/2 \in L$; index 1 at $p = 3$, consistent with the discharged 3-part), and Inputs 1–2 localize it in the B-relative cellular structure of the seven boundary strata (products involving $\overline{\mathcal{M}}_{0,6}$). To our knowledge this is the first computed integral-lattice invariant of a period of $\overline{\mathcal{M}}_{0,8}$ of this kind.

Honest scope. The attribution is rigorous given the measured product. For a measurement-*independent* derivation, subsequent computation has sharpened the gap: the $p = 3$ triviality is now *derived* (incidence entries in $\{0, \pm 1\}$), the arrangement combinatorics is proven innocent under every orientation convention, and a new geometric feature was found — a bi-arrangement non-transversality at $\delta_{24} \cap \delta_{36}$ and $\delta_{14} \cap \delta_{36}$, where δ_{57} becomes a domain facet — the unique locus where the index can be born. What remains is the integral first-differential of the ($\mathbb{Q}$-only in the literature [14]) Orlik–Solomon bicomplex at those two strata. The original statement of the remaining task follows for the record; what remained for a fully measurement-*independent* derivation of the index is one finite computation (the boundary signs of the B-relative cellular complex of the seven strata), which we state as an open task rather than guess: an incorrectly chosen sign convention could manufacture agreement, and we have deliberately not done so. With that computation and hypothesis (H1), the mechanism's account of the constant 12 at the symmetric point — Bernoulli 24 from Lemma 6.1, divided by a computed Betti index 2, with the 3-part exact — would be a theorem. To our knowledge these are the first experimentally measured integral-lattice invariants of periods of $\overline{\mathcal{M}}_{0,8}$; the literature contains the $\mathbb{Q}$-theory [10, 7, 9] and an empirical $\{2, 3\}$-clearing phenomenon for motivic period expansions [11], but—so far as we can determine—no integral-lattice synthesis. The support $\{2, 3\}$ matches the torsion $K_3(\mathbb{Z}) \cong \mathbb{Z}/48$ [12] of the relevant splitting obstruction; we flag this as a consistency, not a derivation.

7 The totally symmetric case: concrete derivation and a support theorem

In the totally symmetric case $\mathbf{a} = (1, \ldots, 1)$ the Barnes representation reduces to the reciprocal-sine kernel with rational part $r(s)r(t)C(s,t)$, $r(s) = \prod_{i=1}^{n}(s+i) / \prod_{j=n+1}^{2n+1}(s+j)^2$, whose partial-fraction data is fully explicit: $B_j = (-1)^n \binom{n+a}{a}\binom{n}{a}^2/n!$ and $A_j = B_j\,(3H_a - H_{n+a} - 2H_{n-a})$ with $a = j - n - 1$ (verified exactly for $n \le 6$). Two consequences, proved in the accompanying notes:

Conjecture 7.1 (support; verified $n \le 8$; large-prime half now proved in Theorem 7.2)**.** *In the totally symmetric case every prime dividing* $\mathrm{den}(P_n)$ *is at most* $2n+1$*, and* $\mathrm{ord}_p \mathrm{den}(P_n) \le 5\,\mathrm{ord}_p d_n$ *for every* $p \ge 5$ *(inequality only: the corresponding equality fails at* $(n,p) = (8,7)$*).* [Added: *for* $p \in (n, 2n]$ *the second bound reads* $\mathrm{ord}_p \mathrm{den}(P_n) = 0$*, and this is now a theorem,*

§7.1.]

Status. An adversarial audit of our own earlier notes found that a claimed unconditional proof of this statement silently invoked the open collapse below, and that even the prime-support bound is open in general (higher sine-kernel Laurent coefficients $\zeta(2k)/\pi^{2k}$ carry Bernoulli numerators with large primes, whose non-contribution is exactly what must be proven). What is proven: the Barnes specialization and the sine-kernel reduction (complete analytic proofs), and the classical partial-fraction/harmonic-clearing integrality inputs. What is open: the general-n "d_n^5-collapse" of the residue sums (the coupling polynomial of the double kernel admits no closed-form reduction to the elementary kernels, which is precisely where an earlier claimed reduction was found to be circular) and the 2-adic attainment. All laws — $12\, d_n^5 P_n \in \mathbb{Z}$, the 2-adic equality $\operatorname{ord}_2 \operatorname{den}(P_n) = 2+5 \operatorname{ord}_2 d_n$, and the $\hat{P}$ companion — are verified exactly for $n \le 8$. Subsequent work (two independently corroborating derivations) organized the collapse into three tiers: a *proven* crude d_{2n}^5-level bound; a *proven* well-poised support law — no $p \ge 5$ excess beyond $p \le n$ is possible, giving theorem-level backing to the support statement's shape — and one named open theorem (a two-variable well-poised $2n \to n$ refinement) plus the Bernoulli constant carrying the $\{2,3\}$-part. Both obstructions are structurally absent from the Catalan sibling of [15].

Moreover the $\{2,3\}$-excess enters *only* through the Laurent coefficients $1, \frac{1}{6}, \frac{1}{90}, \ldots$ of the sine kernel: the constant term P_n picks up exactly one doubled factor of the $\zeta(2)$-coefficient $\frac{1}{6}$, i.e. the factor $12 = 2 \cdot 6$—the concrete avatar of Lemma 6.1—while $\frac{1}{90}$ (which would inject a 5) provably does not contribute, matching Theorem 7.1 and the data. The remaining gap to a full proof of $12\, d_n^5 P_n \in \mathbb{Z}$ is a classical-type d_n^5-collapse bookkeeping for the structured sums above; it is verified exactly for $n \le 5$.

7.1 The central-binomial reduction, and a Frobenius certificate: the large-prime law is a theorem

The strongest unconditional result of this paper closes the large-prime part of the symmetric-case laws. Brown–Zudilin identify the symmetric coefficients with Zudilin's 2002 sequences [2]: $P_n = (-1)^{n+1} p_n / \binom{2n}{n}$, where p_n satisfies Zudilin's denominator theorem $A_n := 2 d_n^5 p_n \in \mathbb{Z}$. Hence

$$12\, d_n^5 P_n \in \mathbb{Z} \quad \Longleftrightarrow \quad \binom{2n}{n} \,\Big|\, 6A_n, \tag{CB}$$

an *extra* divisibility on the numerator that no d_n-denominator estimate can reach at the primes $n < p \le 2n$: these divide $\binom{2n}{n}$ but not d_n. On that interval (CB) reduces (via the p-integrality of the partial-fraction and harmonic data, and $p_n = \widetilde{w}_n v_n - w_n \widetilde{v}_n$) to the congruence (W): $w_n \equiv \widetilde{w}_n \equiv 0 \pmod p$, where $w_n, \widetilde{w}_n$ are the $\zeta(3)$-coefficients of Zudilin's two companion forms $\sum_k R_n(k)$ and $\sum_k -k(k+n)R_n(k)$, with $R_n(k) = (n!)^4 (k+\frac{n}{2}) \prod_1^n (k-j) \prod_1^n (k+n+j) / \prod_0^n (k+j)^6$.

Theorem 7.2 (Frobenius certificate; proved in [3]). *For every $n \ge 1$ and every prime $n < p \le 2n$, $p \ge 5$: $w_n \equiv \widetilde{w}_n \equiv 0 \pmod p$. Consequently* (CB) *holds at every prime of $(n, 2n]$, and* $\operatorname{den}(P_n)$ *contains no prime of $(n, 2n]$—the large-prime half of Conjecture* 7.1 *is unconditionally true.*

Proof sketch. Since $p > n$, all partial-fraction coefficients $a_{i,j}$ of R_n are p-integral and the poles $-0, \ldots, -n$ are distinct in $\mathbb{F}_p$; $w_n \equiv \sum_j \bar{a}_{3,j} \pmod p$. The Frobenius identity $k^p - k = (k+j)^p - (k+j)$ in $\mathbb{F}_p[k]$ makes the *single* polynomial $\varphi = (k^p - k)^2$ satisfy $\varphi \equiv (k+j)^2 \pmod{(k+j)^6}$ at *every* pole simultaneously (here $p \ge 5$), so $\operatorname{Res}_{-j}(\varphi\, \overline{R}_n) = \bar{a}_{3,j}$ for each j. Because $\deg \varphi = 2p$

and $\operatorname{ord}_\infty \overline{R}_n \geq 4n+5$ (the vanishing Laurent coefficients at ∞ reduce mod p), the bound $p \leq 2n$ gives $\varphi\overline{R}_n = O(k^{-5})$; the residue-sum theorem on $\mathbb{P}^1$ then forces $\sum_j \bar{a}_{3,j} = 0$. The companion uses $\psi = -k(k+n)\varphi$ and $p \leq 2n$ once more ($O(k^{-3})$). Full details, the equivalent explicit three-term certificate $c_3 = 1$, $c_{p+2} = -2$, $c_{2p+1} = 1$ on the ∞-expansion relations, and independent exact verification over 420 pairs (n, p), $n \leq 60$, are in [3]. $\square$

The prime window is pinned twice by the proof: $p > n$ for distinct nodes and p-integrality, $2p + \deg Q \leq 4n+3$ for the decay—and the certificate provably cannot exist for $p > 2n+1$, where w_n is generically a unit. The argument uses only the pole orders, the decay, and p-integrality; it is therefore portable to the higher-weight cellular families (poles of order 8 require merely $\varphi \equiv (k+j)^m \bmod (k+j)^8$ with $p + m \geq 8$).

What remains open for the full law $12\, d_n^5 P_n \in \mathbb{Z}$ is *Phase 2*: the primes $p \leq n$, where $\operatorname{ord}_p \binom{2n}{n}$ is governed by Kummer's carry count and interacts with d_n^5 and the harmonic denominators. Two structural remarks sharpen its frontier. First, by Kummer, primes $p \in (\frac{2n}{3}, n]$ contribute no carry ($n = p + r$ with $2r < p$), so $\operatorname{ord}_p \binom{2n}{n} = 0$ and (CB) is *vacuous* there: the first genuinely open band is $p \leq \frac{2n}{3}$, where exactly one extra p-adic order beyond Zudilin's bound is required ($\operatorname{ord}_p \binom{2n}{n} = 1$ on $(\frac{n}{2}, \frac{2n}{3}]$). Second, the exact ledger through $n = 60$ shows 206 zero-slack cases: the required inequality is frequently *tight*, so Phase 2 cannot follow from denominator-size estimates and must track the well-poised structure itself. Phase 2 remains open, and the full law remains a conjecture; the theorem above removes precisely the primes invisible to every classical estimate.

8 Cross-family controls

The same measurement standard applied to the classical corpus shows its correction factors are uniformly *removable*, in contrast to §4: (i) Apéry $\zeta(3)$: here removability is a *theorem*, not merely our finite check: Rhin–Viola [6, Thm. 2.1] prove $d_n^3 a_n \in \mathbb{Z}$ exactly (the genuine 2 rides on the $\zeta(3)$-coefficient parity, not the denominator; the geometric interpretation of such parity vanishing is Dupont's [13, Thm. 1.2]); our $n \leq 50$ verification is a sanity check of their statement. We can now say more: exact residue analysis shows the Rhin–Viola 2 is a *static de Rham period normalization* — $J_0 = 2\zeta(3)$ exactly, because the pole exits the unit cube and $\log(xy) = \log x + \log y$ doubles a boundary sum — separated from our Betti index by three invariants (static vs. growing; constant-term 2-adic excess -2 vs. $+2$; de Rham vs. Betti). Across all measured families, two orthogonal species of constants emerge: static de Rham normalizations ($2\zeta(3)$, $1 \cdot \zeta(5)$, $\frac{75}{4}\zeta(7)$) and Betti elimination costs, paid only by families that eliminate an even period — the Brown–Zudilin family being the unique measured instance. (ii) $\zeta(4)$: $d_n^4 v_n \in \mathbb{Z}$ for $n \leq 5$ in Zudilin's derived series (the factor 2 retained in [5, Théorème 3] is slack there). (iii) Zudilin's asymmetric $\zeta(5), \ldots, \zeta(11)$ forms: computing the five coefficients $z_{j,n}$ exactly for $n \leq 5$ (apparently for the first time), the leading factor 2 retained even in the *conjectured* law [5, eq. (17.1)] is unnecessary, with ~ 30 bits of 2-adic margin; the well-poised parity cancellation and the proven inclusion of [5] serve as engine validations. For (ii) and (iii) removability is genuinely open (Krattenthaler–Rivoal prove their inclusions only "à un facteur 2 près" and could not remove it); our checks are finite verifications, and the structural reason to expect removability there is identifiable but heuristic: the factor 2 in the proofs arises from termwise 2-adic over-counting at half-integer shifts, which the full sums have no reason to respect. What the controls establish is a contrast, not a theorem: in every classical family examined the correction is unneeded in all computed cases, while in the Brown–Zudilin family—the only construction here that performs a $\zeta(2)$-elimination—it is provably needed (the printed values of §3 already require it).

The weight-7 family: first data. For the totally symmetric cellular family on $\overline{\mathcal{M}}_{0,10}$ (rank 4; Brown–Zudilin print $n \le 2$ and call further computation impractical), we established: its $\zeta(2)$-eliminated form provably lies outside the single-sum very-well-poised class (a sign dichotomy: projections carry opposite-sign adjacent zeta coefficients; simultaneous approximations cannot); the cell is dihedrally rigid (trivial stabilizer); and, via the leading-coefficient diagonal construction of McCarthy–Osburn–Straub [16] — the leading coefficient is the diagonal coefficient $q_n = [(x_1 \cdots x_8)^n] \prod_i W_i^n$ for explicit window polynomials W_i attached to the cell — the first new exact values. The construction is validated twice over: it reproduces the printed $q_0, q_1, q_2 = 1, 61, 52921$, and, applied unchanged to the $\overline{\mathcal{M}}_{0,6}$ test cell, reproduces the classical Apéry $\zeta(3)$ numbers $1, 5, 73, 1445, 33001, 819005$. The new terms begin

$$q_3 = 94357501, \quad q_4 = 235634763001, \quad q_5 = 715362962769061,$$

with $q_0, \ldots, q_{30}$ (the last a 108-digit integer) computed exactly and archived with the code; the growth ratio approaches $\lambda_{\max} \approx 5.8 \times 10^3$, the family's leading asymptotic rate. The recurrence itself — hence the characteristic polynomial, the propagation test for the constant-term ladder $P_0 = 0$, $P_1 = 220$, $P_2 = 6021219/32$, and the family's 3-adic elimination-cost test — is reduced to one standard creative-telescoping computation on the diagonal (in progress at the time of writing; thirty-one terms exclude operators of order ≤ 8 at moderate degree, consistent with a high-degree order-4 operator as in the weight-5 family).

9 The worthiness supremum, and what the arithmetic cannot do

For completeness we record the analytic side. Implementing the worthiness exponent $\gamma(\mathbf{a})$ of [1] (validated to 8 decimals against all three published values, including every printed intermediate), a global search of the 7-dimensional projective cone —steepest ascent from the published point on integer and half-integer lattices, 120 random multi-starts, 60 basin hops, and an exhaustive scan of all 31,710 projectively distinct arithmetic-progression points to height 160—finds the supremum

$$\sup_{\mathbf{a}} \gamma \; = \; 0.86597135\ldots,$$

attained exactly at the published point $\mathbf{a} = (8, 16, 10, 15, 12, 16, 18, 13)$, whose symmetric coordinates $(20.5; 3.5, 4.5, \ldots, 9.5)$ form an arithmetic progression. Combined with the bounded-slack measurements of §4, the construction's arithmetic is now understood to bounded factors, and none of it moves the exponential budget: within this family the barrier to $\gamma > 1$ is structural.

10 Limitations and outlook

All sharpness and slack statements are finite verifications under the certification protocol of §2; Conjecture 4.1 is falsifiable by a single future cell. The mechanism of §6 is proved only in the elementary Lemma 6.1 plus the conditional reduction; hypothesis (H2) is one finite lattice computation (torsion-freeness of $H^*(\overline{\mathcal{M}}_{0,n}, \mathbb{Z})$ [17] should make it tractable), and (H1) is the integral form of the Barnes bookkeeping sketched in [1]. The measurement pipeline itself is portable, and we suggest its use as an *arithmetic screen* for any future candidate constructions: the true denominators of a proposed family can now be measured in hours rather than conjectured. In a companion development, the full theorem of [4] has been formalized in Lean 4/Mathlib in the form: at least one of $\zeta(5), \zeta(7), \ldots, \zeta(33)$ is irrational (the $s = 33$ variant via Hanson's elementary $d_n \leq 3^n$, formalized from scratch since Mathlib currently lacks the prime number theorem); sorry-free, kernel axiom set [`propext`, `Classical.choice`, `Quot.sound`]. Details appear in the companion paper accompanying the formalization.

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